

**Project Title :
Used Car Price Prediction Using Regression
Model**

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1. Abstract

Accurate valuation of pre-owned vehicles is essential for optimizing transactions in the secondary automotive market. This study develops a machine learning framework to predict used car prices by leveraging various advanced regression models. A comprehensive dataset comprising historical vehicle sales, technical specifications, mileage, and brand reputation is utilized for model training. The proposed framework aims to enhance pricing accuracy and support data-driven decision-making for buyers, sellers, and automotive businesses in the pre-owned vehicle market.

2. Introduction

The rapid proliferation of the secondary automotive market has necessitated more transparent and reliable pricing mechanisms. Traditional valuation methods often rely on manual assessment and market experience, which can be subjective and inconsistent. Therefore, there is a growing need for an automated, accurate, and data-driven approach to used car price estimation.

This project focuses on developing a Machine Learning (ML) model for predicting used car prices based on historical vehicle data and technical specifications. By analyzing historical sales records and vehicle attributes, the proposed system aims to estimate the market value of a used car with high accuracy.

The project utilizes various technologies and tools including Python as the programming language, Pandas and NumPy for data preprocessing and analysis, Matplotlib and Scikit-learn for implementing machine learning algorithms. Advanced regression techniques such as Linear Regression, Random Forest Regression was employed and compared to identify the most effective prediction model.

The prerequisite was obtained from the two weeks training phase, before the commencement of this project. This training program covered topics on-

- Python Coding
- Machine Learning (Overview along with development)
- Large-Language Models (LLMs)
- Communication Skills

3. Project Objective

- To develop a machine learning-based regression model capable of accurately predicting the selling price of used cars using vehicle attributes and historical market data.
- To identify and analyze the key factors influencing used car prices, such as mileage, age, brand, fuel type, transmission, and engine specifications.
- To preprocess and transform raw vehicle data through data cleaning, feature engineering, and encoding techniques to improve model performance and reliability.

- To implement and compare multiple regression algorithms, such as Linear Regression, Decision Tree Regression, Random Forest Regression, and Gradient Boosting Regression, in order to determine the most effective prediction model.
- To evaluate the predictive performance of the developed models using statistical metrics such as Mean Absolute Error (MAE) and R-squared (R^2) score, and select the optimal model for practical deployment.

4. Methodology

The methodology adopted for the Used Car Price Prediction project consists of a systematic sequence of steps designed to develop an accurate and reliable machine learning model for estimating the market value of pre-owned vehicles. The process began with **data collection**, where a dataset containing information such as vehicle brand, model, year of manufacture, mileage, fuel type, transmission type, engine capacity, ownership history, and selling price was obtained from my mentor.

The collected data was then subjected to **data preprocessing and cleaning**. During this stage missing values are handled appropriately. Outliers that may adversely affect model performance are detected and treated. Categorical features are converted into numerical representations using encoding techniques.

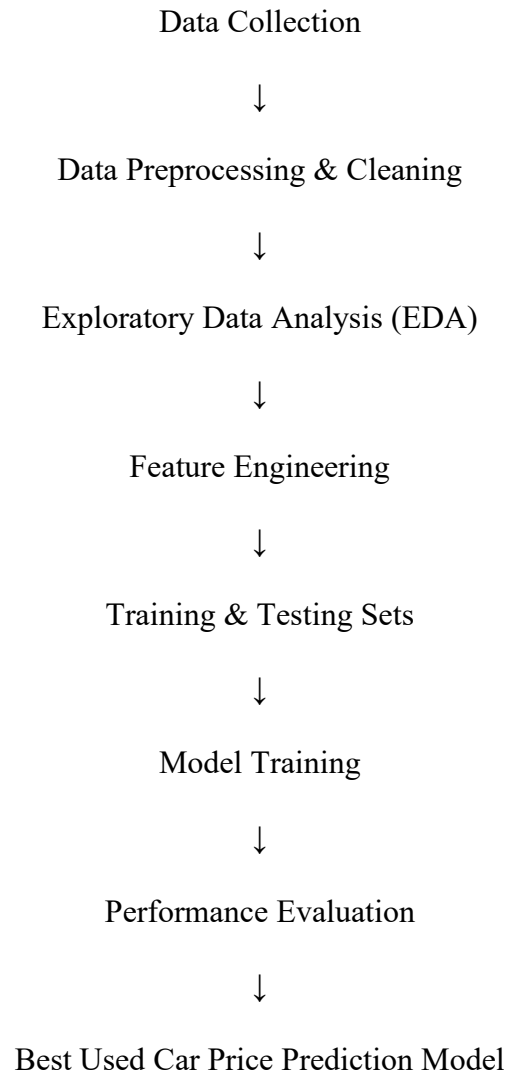
Following preprocessing, **Exploratory Data Analysis (EDA)** was performed to understand the underlying patterns and relationships within the dataset. Correlation analysis helps determine the strength of relationships between independent variables and the target variable.

The next phase involves **feature selection and feature engineering**. Relevant features that contribute significantly to price prediction are selected, while redundant or irrelevant attributes are removed.

After feature preparation, the dataset was divided into **training and testing sets**. 80% of the data is allocated for training the model, while the remaining 20% data is reserved for evaluating its performance. Regression algorithms, including Linear Regression and Random Forest Regression were then implemented and trained using the prepared dataset.

Once training is completed, the models underwent **performance evaluation** using metrics such as Mean Absolute Error (MAE), Mean Squared Error (MSE), and R-squared (R^2) score. These metrics provide insights into prediction accuracy and model reliability. A comparative analysis of all implemented regression models is conducted to identify the algorithm that delivers the best performance.

The entire process can be simply demonstrated as a flowchart -



5. Data Analysis and Results

The results from our Machine Learning Model are:

- R^2 Score: 0.7286497669639422
- Mean Squared Error (MSE): 319,333,182.34
- Mean Absolute Error (MAE): 9,462.83

The R^2 score indicates that the model explains approximately:

$$0.72865 \times 100 \approx 72.86\%$$

of the variance in used car prices.

The model obtained a Mean Absolute Error (MAE) of \$9,462.83 and a Mean Squared Error (MSE) of 319,333,182.34, demonstrating a reasonably good predictive capability for estimating used car prices.

6. Conclusion

- The dataset was successfully cleaned and prepared through preprocessing techniques, improving data quality and model reliability.
- Exploratory Data Analysis identified important relationships between vehicle characteristics and selling price.
- Feature engineering and encoding techniques enabled the model to utilize both numerical and categorical data effectively.
- The predictive performance can be enhanced by incorporating additional features such as service history, accident records, vehicle condition ratings, market demand trends, and geographical location. Advanced machine learning algorithms and hyperparameter optimization techniques may further improve prediction accuracy.

7. Appendices

1. [IDEAS-TIH-ISI-Kolkata/15 Used Car Price Prediction Spring 2026.ipynb at main • sankalpaghosh29-rgb/IDEAS-TIH-ISI-Kolkata](#)
2. [sankalpaghosh29-rgb/IDEAS-TIH-ISI-Kolkata](#)